

Monte Carlo Simulation for Tail Risk

Calibration, Simulation, and the Gap with Reality

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Abstract

These are working notes on understanding *tail risk* through Monte Carlo simulation. The guiding question is the relationship between **calibration** (fitting a model to data we have seen) and **reality** (the losses the world actually delivers). I build a dataset from a known fat-tailed, volatility-clustering process, calibrate three risk models to it (Normal, Student- t , and an Extreme-Value/Generalized-Pareto tail model), estimate Value-at-Risk and Expected Shortfall by Monte Carlo, and then *backtest* each calibration against the realised series. The recurring lesson: a model can fit the body of the data and still be dangerously wrong in the tail, and calibration only ever recovers the model you assumed — not necessarily the process that generated the world.

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1 The question

A risk number is a claim about the future stated as a probability. When we say “the one-day 99% Value-at-Risk is 3.2%,” we are asserting that a loss worse than 3.2% should happen on about one day in a hundred. Two separate things must go right for that claim to hold:

1. **Calibration** — we picked model parameters that match the data.
2. **Specification** — the model *family* is capable of describing reality in the first place.

Calibration is an optimisation problem and always “succeeds”: it returns the best parameters *within the family you chose*. Whether those parameters say anything true about the world is a different question, answered only by confronting the model with out-of-sample reality. These notes make that distinction concrete and measurable.

2 Tail risk: VaR and Expected Shortfall

Let L denote the loss over the risk horizon (here, one day), so that $L = -r$ for a log-return r . Fix a confidence level $\alpha \in (0, 1)$ (e.g. 0.99).

Definition 1 (Value-at-Risk). $\text{VaR}_\alpha(L)$ is the α -quantile of the loss distribution,

$$\text{VaR}_\alpha(L) = \inf\{\ell \in \mathbb{R} : \mathbb{P}(L \leq \ell) \geq \alpha\}.$$

VaR answers “how bad is a bad day, at the α threshold?” It says nothing about *how much worse* the rare day beyond the threshold can be, and it is not sub-additive — diversification can appear to *increase* it. The fix is the conditional average of the tail:

Definition 2 (Expected Shortfall).

$$\text{ES}_\alpha(L) = \frac{1}{1-\alpha} \int_\alpha^1 \text{VaR}_u(L) \, du = \mathbb{E}[L \mid L \geq \text{VaR}_\alpha(L)] \quad (\text{for continuous } L).$$

ES (a.k.a. CVaR or TVaR) is *coherent*: monotone, translation invariant, positively homogeneous, and sub-additive. It is the average loss on the worst $(1-\alpha)$ fraction of days, and it is the quantity post-2016 Basel rules put at the centre of market-risk capital. Both VaR and ES depend entirely on the *shape of the tail* — which is exactly the part of the distribution that data is sparsest about.

3 The data and its stylised facts

To study calibration against a known truth, I generate 3,000 daily log-returns (≈ 12 trading years) from a GARCH(1, 1) process with standardised Student- t innovations:

$$r_t = \mu + \sigma_t z_t, \quad \sigma_t^2 = \omega + a \varepsilon_{t-1}^2 + b \sigma_{t-1}^2, \quad z_t \sim t_\nu(\text{unit variance}),$$

with true parameters $\mu = 5 \times 10^{-4}$, $\omega = 2 \times 10^{-6}$, $a = 0.08$, $b = 0.90$, and $\nu = 4$. This process is the “reality” in the experiment. It reproduces the two stylised facts that make tail risk hard:

Fat tails. The realised series has excess kurtosis ≈ 23.7 (Gaussian = 0): extreme moves are vastly more frequent than a Normal allows.

Volatility clustering. Variance is persistent ($a + b = 0.98$): turbulent days follow turbulent days. Figure 1 shows the bursts.

A Normal QQ-plot (Figure 2) is the quickest diagnostic: if the data were Gaussian the points would lie on the line; instead the tails peel away, the signature of a heavy-tailed law.

4 Calibration: three models

I fit three models to the loss series $L = -r$ by maximum likelihood, in order of increasing tail-awareness.

Normal. Two parameters (μ, σ) , fit by the sample mean and standard deviation. Convenient, closed-form everything, and thin-tailed by construction — the null hypothesis we expect to fail.

Student- t . Adds a degrees-of-freedom parameter ν that fattens the tails; as $\nu \rightarrow \infty$ it collapses to the Normal. Fit by MLE over $(\nu, \text{loc}, \text{scale})$.

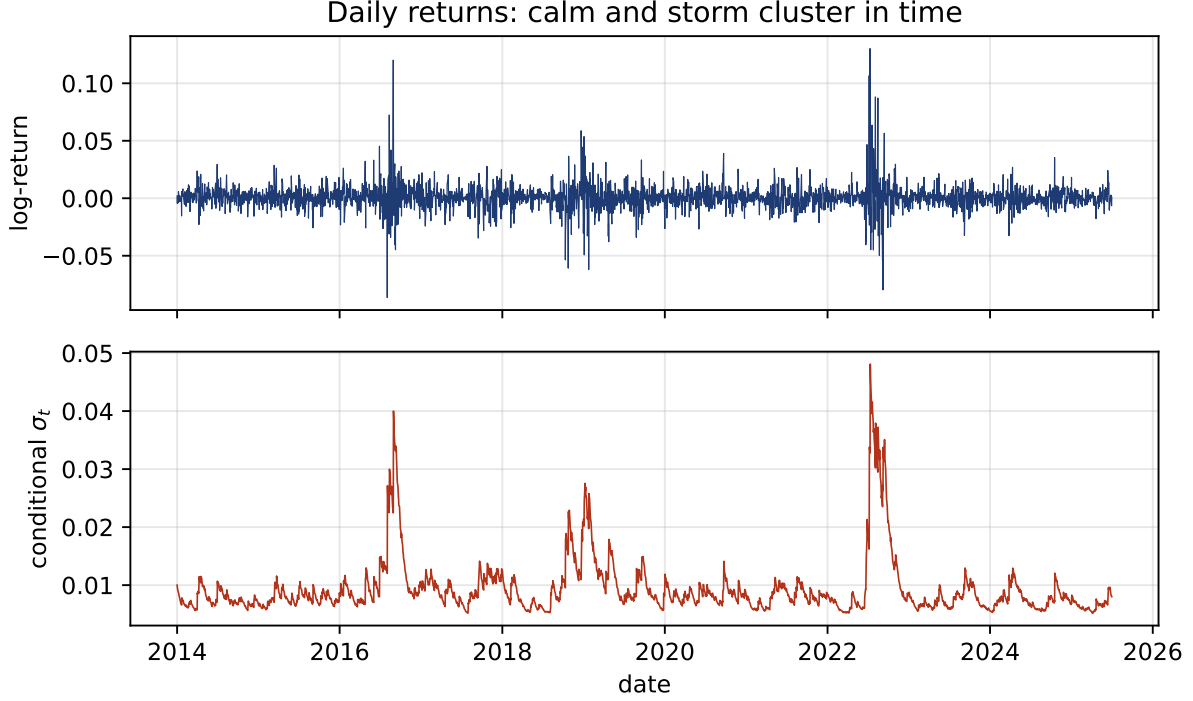


Figure 1: Top: simulated daily returns. Bottom: the latent conditional volatility σ_t . Large moves and high volatility cluster — they are not sprinkled uniformly through time.

Extreme Value Theory (Peaks-Over-Threshold). Instead of modelling the whole distribution, EVT models *only* the tail. The Pickands–Balkema–de Haan theorem says that for a high threshold u , the exceedances $L - u \mid L > u$ converge to a Generalized Pareto Distribution (GPD):

$$G_{\xi, \beta}(y) = 1 - \left(1 + \frac{\xi y}{\beta}\right)^{-1/\xi}, \quad y > 0.$$

The shape $\xi > 0$ signals a heavy (power-law) tail. I set u at the 90% quantile, leaving 300 exceedances, and fit (ξ, β) by MLE.

Table 1: Calibrated parameters (loss space).

Model	Parameters		
Normal	$\mu = -0.00063$	$\sigma = 0.01080$	
Student- t	$\nu = 2.21$	loc= -0.00062	scale= 0.00536
GPD (POT)	$u = 0.0094$	$\xi = +0.260$	$\beta = 0.00598$

A first surprise — calibration $\neq$ truth. The true *conditional* innovation has $\nu = 4$, yet the calibrated i.i.d. Student- t reports $\nu \approx 2.21$. This is not an error; it is the central lesson appearing early. Mixing many days of *different* volatilities (the GARCH effect) produces an *unconditional* distribution with much heavier tails than any single conditional t_4 . Fitting an i.i.d. model to a clustered reality forces ν down to absorb that extra kurtosis. The number that “best fits” is faithful to the *assumed family*, not to the mechanism that generated the data. Figure 3 shows the cost of the thin-tailed assumption directly.

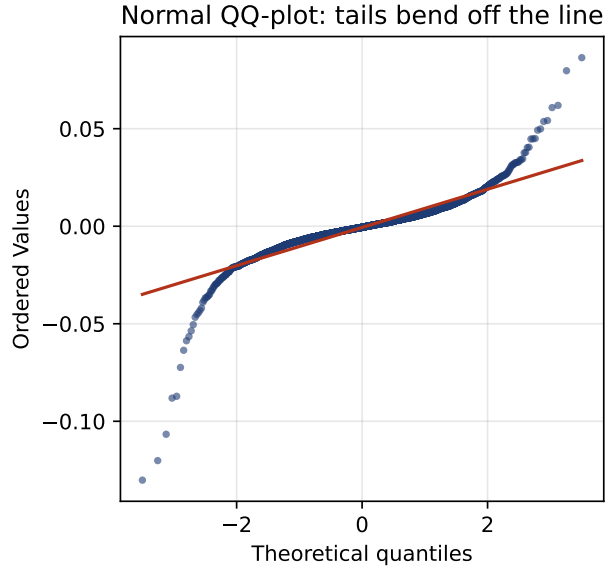


Figure 2: Normal QQ-plot of the losses. The systematic departure at both ends is the visual fingerprint of fat tails.

5 Monte Carlo simulation

For the Normal and Student- t fits, VaR and ES have closed forms. Real portfolios — options, path-dependence, many coupled risk factors — usually do not. Monte Carlo is the general method:

1. Draw N i.i.d. losses $L^{(1)}, \dots, L^{(N)}$ from the calibrated model.
2. Estimate $\widehat{\text{VaR}}_\alpha$ as the empirical α -quantile.
3. Estimate $\widehat{\text{ES}}_\alpha$ as the mean of the draws $\geq \widehat{\text{VaR}}_\alpha$.

The estimator carries *sampling noise* of order $1/\sqrt{N}$. For a quantile, the asymptotic standard error is

$$\text{s.e.}(\widehat{\text{VaR}}_\alpha) \approx \frac{1}{f(\text{VaR}_\alpha)} \sqrt{\frac{\alpha(1-\alpha)}{N}},$$

which blows up as $\alpha \rightarrow 1$ (the density f in the tail is tiny). Tail quantiles are intrinsically the hardest to simulate — precisely why variance reduction and EVT exist. Figure 4 shows convergence to the analytic value.

Table 2: Monte Carlo ($N = 200,000$) vs. closed form, calibrated Student- t . Agreement here validates the *simulation*; it says nothing about whether the model is *right*.

	$\alpha = 95\%$		$\alpha = 99\%$	
	VaR	ES	VaR	ES
Monte Carlo	0.0141	0.0279	0.0326	0.0597
Analytic	0.0141	0.0280	0.0323	0.0603

6 Calibration versus reality: backtesting

This is the heart of the matter. Each calibrated model emits a VaR. Reality — the 3,000 realised days — emits actual losses. A *breach* is a day on which the realised loss exceeds the model's VaR. At level α we expect breaches on a $(1 - \alpha)$ fraction of days. We test that with the Kupiec

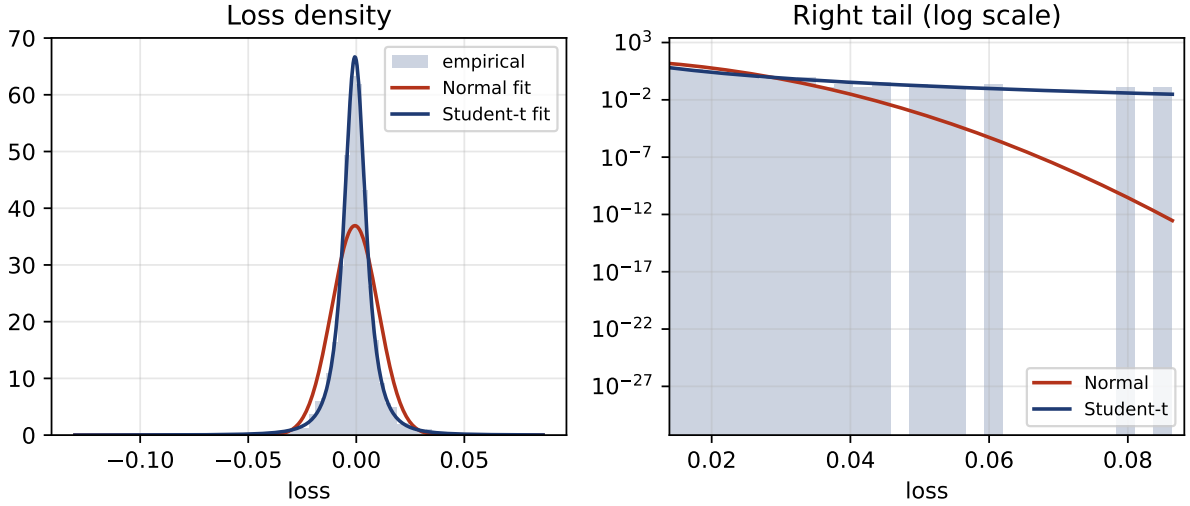


Figure 3: Left: loss histogram with the Normal and Student- t fits. Right: the same right tail on a log scale. The Normal (red) plunges to zero; the Student- t (navy) keeps mass out where the real losses live.

proportion-of-failures likelihood-ratio statistic

$$LR_{\text{POF}} = -2 \ln \frac{(1-p)^{n-x} p^x}{(1-\hat{\pi})^{n-x} \hat{\pi}^x} \xrightarrow{H_0} \chi_1^2, \quad \hat{\pi} = \frac{x}{n}, \quad p = 1 - \alpha,$$

where x is the breach count over n days. A small p -value means the observed breach rate is inconsistent with the claimed level.

Table 3: Backtest over $n = 3,000$ realised days. The Normal model is **rejected** at both levels; the fat-tailed models pass.

α	Model	VaR	Breaches	Expected	Kupiec p	Verdict
95%	Normal	0.0171	100	150.0	0.000	REJECT
	Student- t	0.0141	148	150.0	0.867	PASS
	GPD/EVT	0.0139	150	150.0	1.000	PASS
99%	Normal	0.0245	42	30.0	0.038	REJECT
	Student- t	0.0323	23	30.0	0.180	PASS
	GPD/EVT	0.0283	29	30.0	0.854	PASS

Read the Normal row carefully, because it shows *two opposite failures of the same model*:

- At 95% it is *too conservative* (100 breaches vs. 150): its wide “body” overstates ordinary risk and wastes capital.
- At 99% it is *too reckless* (42 breaches vs. 30): its thin tail understates extreme risk — the failure that bankrupts you.

A single mis-specified family is simultaneously charging too much for calm and too little for catastrophe. Figure 5 marks where reality broke through the 99% lines.

7 What this says about calibration and reality

1. **Fitting the body is not fitting the tail.** The Normal nails the mean and variance and still fails where it matters. Goodness-of-fit in the centre is no evidence about the extremes.

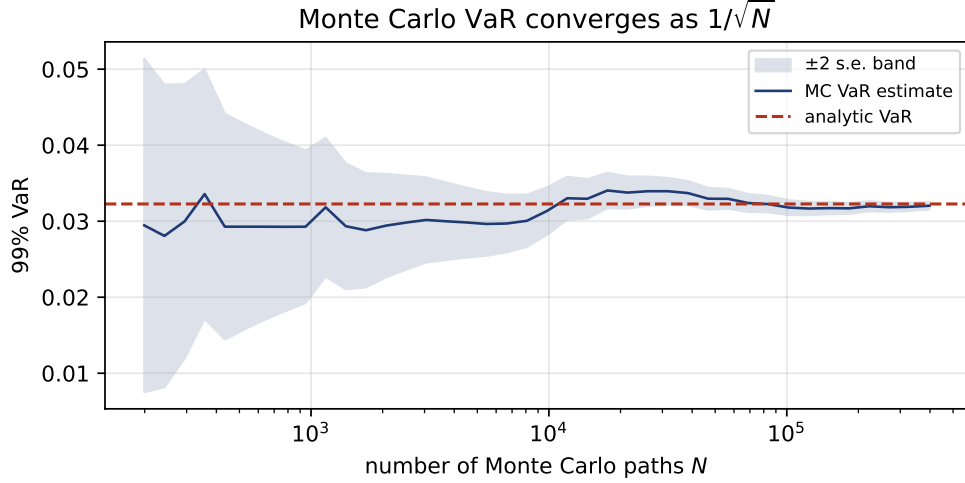


Figure 4: Monte Carlo 99% VaR versus number of paths N , with a ± 2 standard-error band, converging to the closed-form value (dashed).

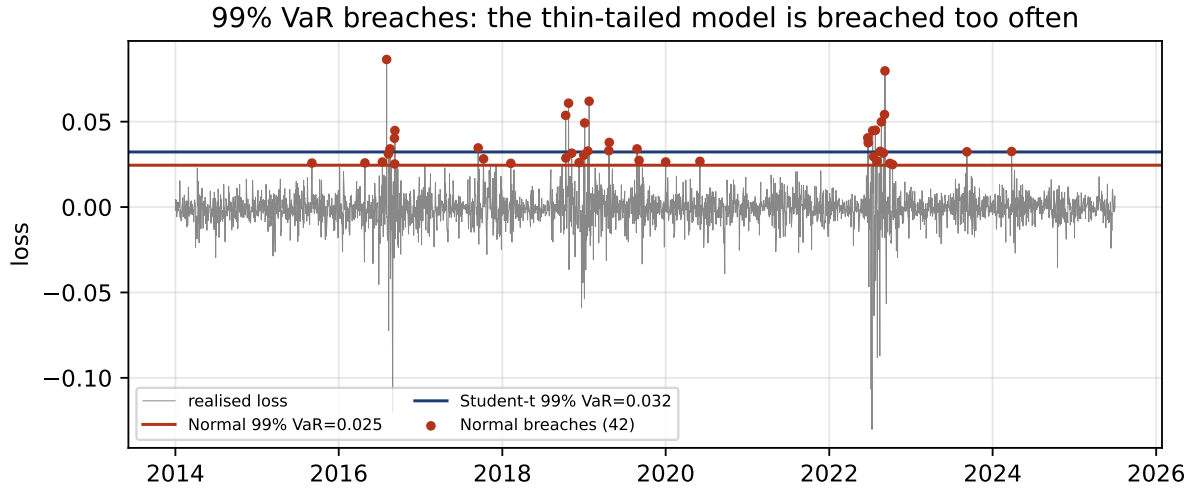


Figure 5: Realised losses with each model's 99% VaR. Red dots are days that breached the Normal VaR — visibly clustered, another reminder that breaches arrive together, not independently.

2. **Calibration recovers the model, not the truth.** The fitted $\nu = 2.21$ is the right answer to the wrong question; the true conditional ν was 4. An i.i.d. lens cannot see a clustered world, so it relabels persistence as fatter tails.
3. **Simulation accuracy $\neq$ model accuracy.** Table 2 shows Monte Carlo reproducing the analytic VaR to four decimals. That only proves the *arithmetic* is right. Backtesting (Table 3) is what tests the *model*.
4. **Estimate ES, and prefer EVT in the tail.** ES integrates the whole tail and is coherent; EVT models the tail on its own terms and here tracked reality most closely (breach rates 5.00% and 0.97%).
5. **Breaches cluster.** Both the data and Figure 5 show exceedances arriving in bursts, violating the independence that simple VaR assumes — which is why conditional (e.g. GARCH-filtered) and dynamic-EVT approaches exist.

The map is not the territory. A calibrated model is a map drawn from the ground we have already walked. Tail risk lives in the ground we have not. The discipline is to keep asking how the map could be wrong — by backtesting, by stressing parameters, and by choosing families (Student- t , EVT) honest about how heavy reality's tails can be.

A Reproducing these notes

From `src/` run, in order:

```
python3 generate_data.py    # writes data/returns.csv (the "reality")
python3 calibrate.py        # fits Normal / Student-t / GPD -> results.json
python3 simulate.py         # Monte Carlo VaR & ES
python3 backtest.py         # Kupiec backtest vs realised losses
python3 plots.py            # regenerates every figure in this document
```

All randomness is seeded in `config.py`, so the numbers quoted here reproduce exactly. Change `TRUE_NU`, `TRUE_BETA`, or the confidence levels there to run your own calibration-versus-reality experiments.